Chapter 2 - Variables and Data Types

What are Python Variables?

A variable is a name that refers to a value stored in memory. You create a variable by assigning a value with the = operator. Python is dynamically typed, so the same name can later refer to a value of a different type. Variable names can contain letters, digits and underscores, cannot start with a digit, and are case-sensitive. The convention is snake_case, for example student_name.

```
age = 20            # int
price = 99.5        # float
name = "Ayesha"      # str
is_student = True   # bool
nothing = None      # NoneType
print(type(age))    # <class 'int'>
```

Basic Data Types

- int: whole numbers such as 10 or -3.
- float: decimal numbers such as 3.14.
- str: text inside quotes, such as 'Python'.
- bool: True or False.
- NoneType: the special value None meaning 'no value'.

Type conversion (casting) changes a value from one type to another using int(), float(), str() and bool(). For example int('42') gives the number 42. F-strings make it easy to put variables inside text.

```
score = int("85")
print(f"{name} scored {score} marks")
```

Collection Data Types

- list: an ordered, mutable collection written with square brackets [1, 2, 3].
- tuple: an ordered, immutable collection written with parentheses (1, 2, 3).
- dict: key-value pairs written with curly braces {'name': 'Ali'}.
- set: an unordered collection of unique items {1, 2, 3}.

Difference Between a List and a Tuple

The main difference is mutability. A list is mutable, which means you can add, remove or change its items after creation. A tuple is immutable, so once it is created it cannot be changed. Lists use square brackets and tuples use parentheses. Because tuples are fixed, they are slightly faster, use less memory and can be used as dictionary keys. Use a list for data that changes (a shopping cart) and a tuple for fixed data (coordinates or days of the week).

```
fruits = ["apple", "banana"]
fruits.append("mango")     # works: lists are mutable
fruits[0] = "kiwi"          # works

point = (10, 20)
# point[0] = 5               # TypeError: tuples are immutable
```